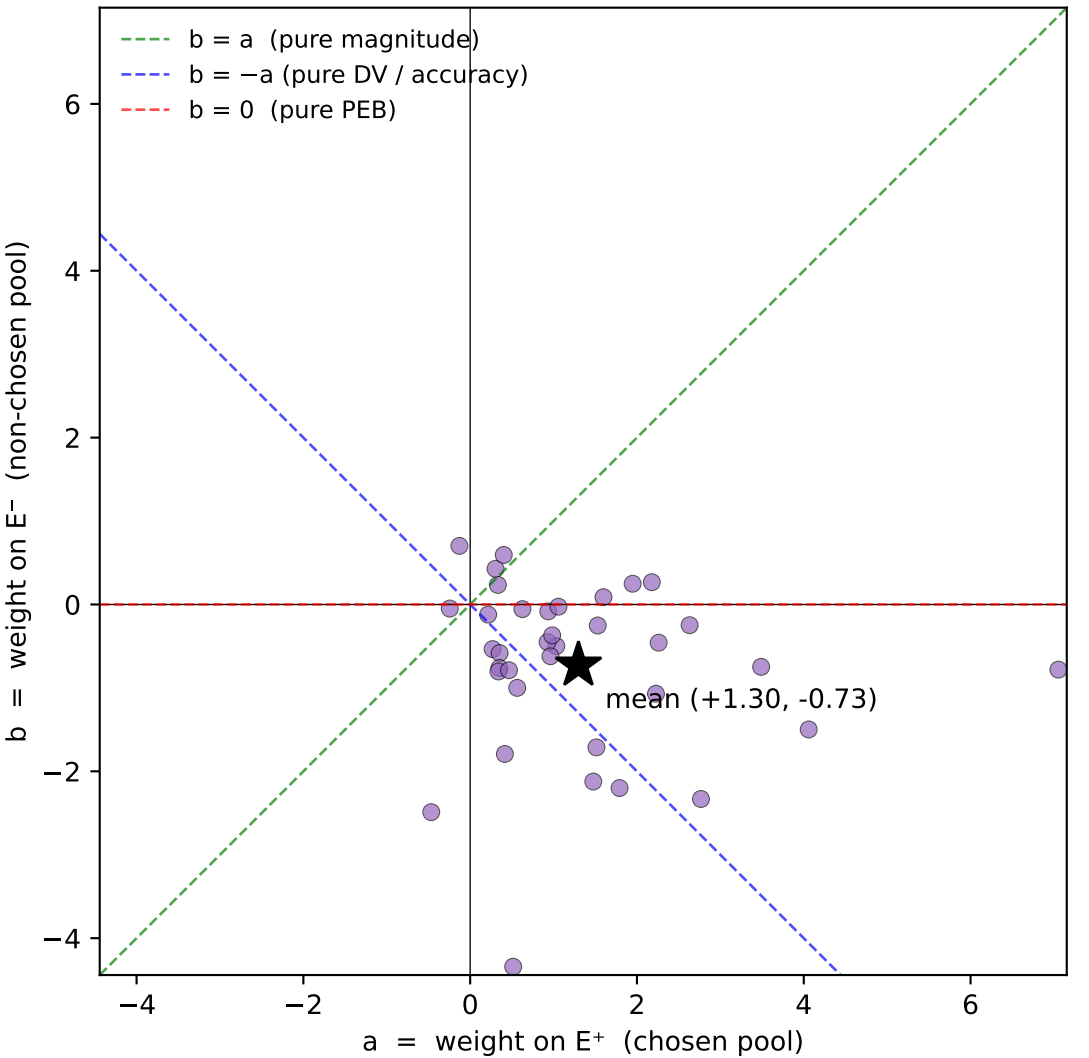
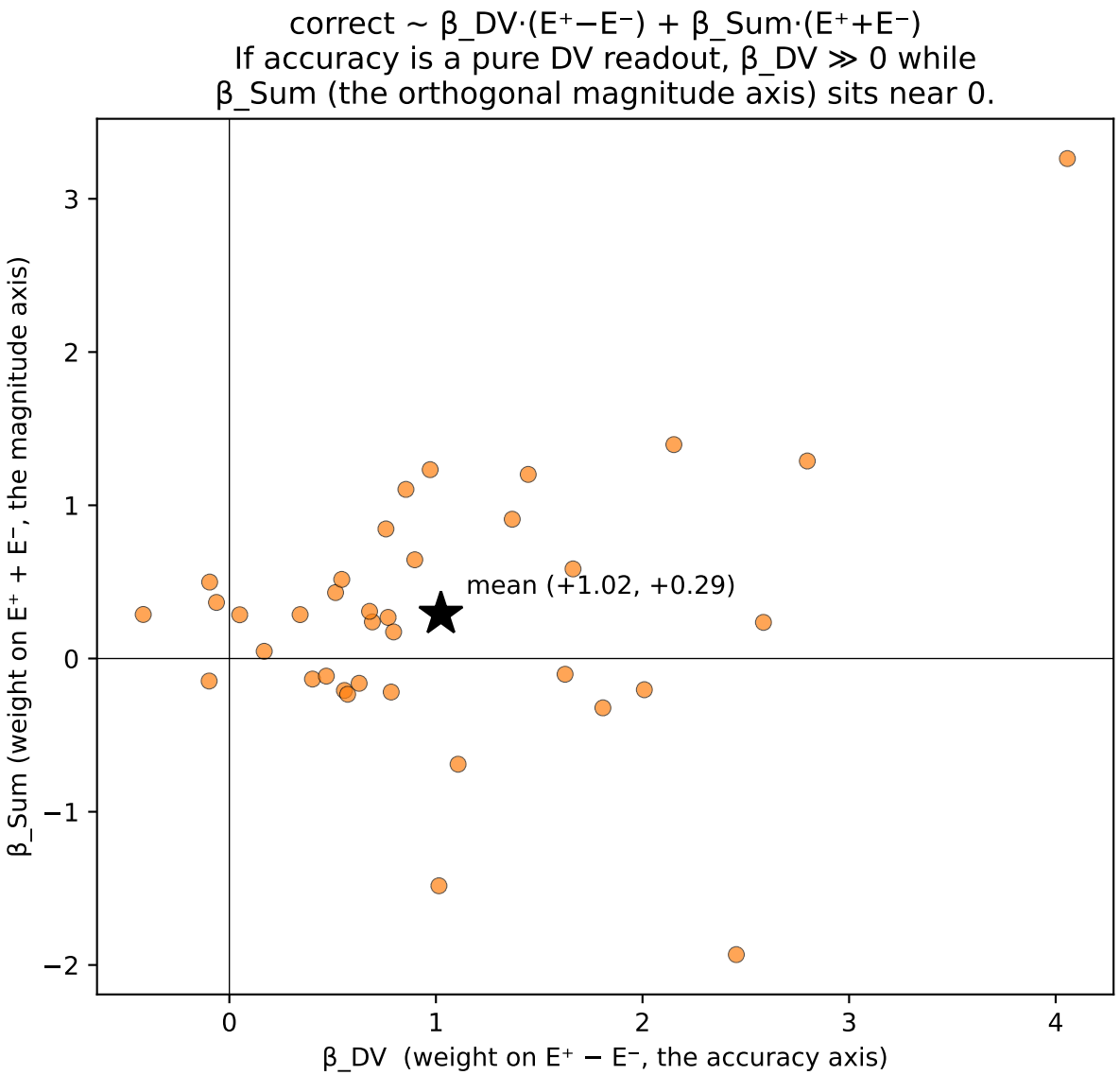
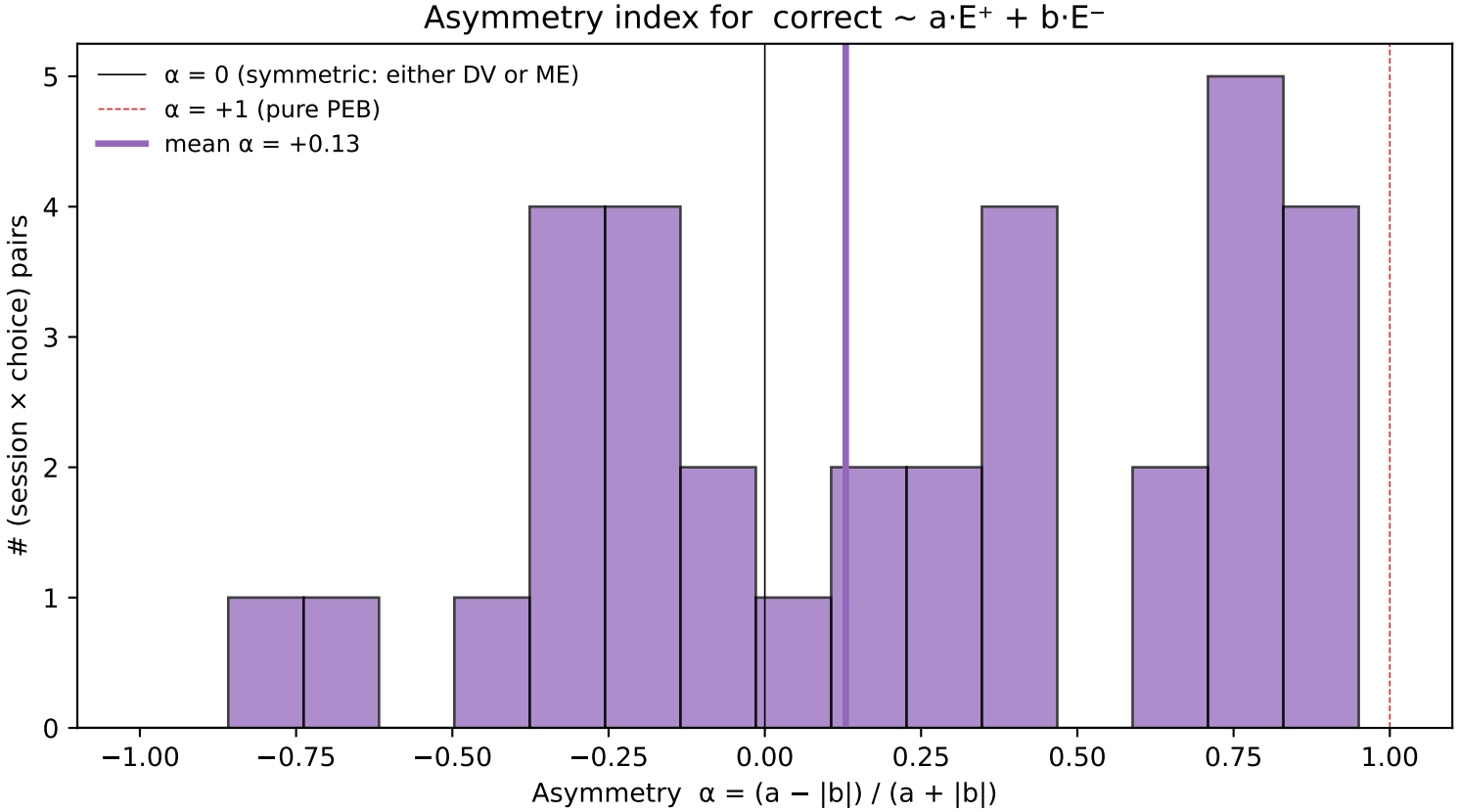

correct ~ a·E⁺ + b·E⁻ (one dot per session × choice)
Where in (a, b) space do the data sit?
b = a (pure magnitude)
b = −a (pure DV / accuracy)
b = 0 (pure PEB)
mean (+1.30, -0.73)
a = weight on E⁺ (chosen pool)
b = weight on E⁻ (non-chosen pool)
6
4
2
0
−2
−4
−4
−2
0
2
4
6


correct ~ β_DV·(E⁺−E⁻) + β_Sum·(E⁺+E⁻)
If accuracy is a pure DV readout, β_DV ≫ 0 while
β_Sum (the orthogonal magnitude axis) sits near 0.
mean (+1.02, +0.29)
β_Sum (weight on E⁺ + E⁻, the magnitude axis)
β_DV (weight on E⁺ − E⁻, the accuracy axis)
3
2
1
0
−1
−2
0
1
2
3
4


Asymmetry index for correct ~ a·E⁺ + b·E⁻
α = 0 (symmetric: either DV or ME)
α = +1 (pure PEB)
mean α = +0.13
(session × choice) pairs
0
1
2
3
4
5
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
Asymmetry α = (a − |b|) / (a + |b|)

```
PEB vs ME — ACCURACY target (18 sessions, 36 pairs after pref labels)

correct ~ a · E⁺ + b · E⁻   (logistic, E± z-scored within sess×choice)
  a (chosen pool)      mean = +1.298   p vs 0 = 6.02e-09
  b (non-chosen pool)  mean = -0.729   p vs 0 = 3.33e-05
  Wilcoxon (a+b ≠ 0):  mean = +0.570   p = 1.10e-02

Asymmetry α = (a − |b|)/(a + |b|):  mean = +0.129
  α ≈ +1 → pure PEB,   α ≈ 0 → symmetric (DV if b<0, ME if b>0)

Same model in DV / Sum basis:
  β_DV   (accuracy axis)  = +1.024   p = 6.02e-09
  β_Sum  (magnitude axis) = +0.291   p = 1.05e-02

Interpretation rules:
  pure DV / accuracy           a > 0,  b = −a   (sum a+b = 0)
  positive-evidence bias       a > 0,  b ≈ 0    (α near +1)
  magnitude / gain effect      a = b > 0        (α near 0,  with b > 0)

Reading off the data:
  Sign of a:   + (positive)
  Sign of b:   − (negative)
  Sign of (a+b):  +
  (a+b ≈ 0 and b < 0  →  accuracy ≈ pure DV, as expected)
```